

PROJECT BASED LEARNING

SMART PARKING SYSTEM

The project report submitted in partial fulfillment of the requirement of the degree

of

BACHELOR OF TECHNOLOGY

In

ELECTRONICS AND COMMUNICATION ENGINEERING

By

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

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Certificate

This is to certify that the project report entitled "Smart parking system " submitted by Ankit Kumar (202200158) to Sikkim Manipal Institute Of Technology, Sikkim in partial fulfillment for the award of a degree of Bachelor of Technology in Electronics And Communication Engineering, is a Bonafide record of the project work carried out by him under my guidance

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ABSTRACT

This project is to set up a smart parking system. According to primary investigation, nowadays maximum people want a smart parking system for hassle free parking with digital assistant. The IR sensor Range Detection Sensor is utilized with Arduino to indicate the empty slot. By measuring the distance using IR sensor drivers are able to find the empty slot in parking to park the car and help the driver to find the slot easily and reduce the searching time. As the parking place is found to be empty it is detected using IR sensors. We achieved this by programming the sensors and Arduino. This sensor and Arduino Microcontroller based technological solution can prevent inadequate parking, take payment digitally and save time and manpower. This can be a sustainable solution in a cheaper cost. The discussed systems will be able to reduce the problems which are arising due to unavailability of a reliable, efficient and modern parking system, while the economic analysis technique will help in analysing the projects' feasibility.

Smart car parking project aims at providing a confusion free and easy parking. This project helps the drivers of the cars to park their vehicles with minimum wastage of time with accurate information of the availability of the space to park. It includes an Arduino Uno as the microcontroller unit to which the servo motors, LCD display, IR sensors are interfaced. The LCD displays the availability of the space, the ultrasonic sensors keep the check of the number of cars entering and exiting the parking space. The ultrasonic sensors detect the availability of the parking space.

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1. INTRODUCTION

1.1 Background of study and motivation:

Parking is a major problem in our country. Drivers in our country cannot properly park their vehicle in the parking slot because of improper parking system. Sometimes when they do not find any parking space, they park their vehicle in the roadside and as a result roads become so narrow and occurs traffic jam. Also, sometimes it seems that people start fighting for parking slot in the parking lot. Because from outside people cannot understand whether there is any parking slot is available or not. It also wastes our lot of time. Till now no smart technology exists to solve this problem. This project will change and standardize the parking system for a parking lot.

An automated car parking system is a process through which car parking can be done more efficiently and easily than manual method. The system will provide the user better services.

The system counts the number of cars in the garage and checks if there's any vacancy. There's an entry and exit path. When vehicle enters, the display shows the number of cars inside. When any vehicle leaves, the count decreases and shown on display. If the garage is full. The display will show a message regarding that. This whole process includes the use of Arduino, Display and sensor. The sensor detects whether the vehicle is entering or leaving. The report then showed on display.

1.2 Project Objectives:

The objectives to be achieved are:

- Providing safe and secure parking spots within a limited space.
- Developing a sophisticated automated parking system that lowers workforce, and traffic congestion and save time.
- Automatically keeping count and monitoring total entered and exited vehicle and calculate free available slots.

2. LITERATURE SURVEY

Sl no.	Paper name	Authors	Paper details	Relevance to work
1.	Smart Parking Assistance using Arduino	S.K.Satyanarayan, A.Akhil, G.Padmini	International Research Journal of Modernization in Engineering Technology and Science / Volume:04/Issue:03/March-2022	Overall idea of the issues in smart parking
2.	Smart car parking using Arduino uno	Alimul rajee , israt jahan maria	Published on research gate On June 2022	Design implementation idea
3.	Smart parking system with LabVIEW simulator	Rahmad hidyat , Herawati , adndrew ghea mahardia	Journal of physics conference series 239(1):012045 December 2022	Simulation ideas

3. Methodology and Modelling

3.1 Introduction:

This chapter will cover the details explanation of methodology that is being used to make this project complete and working well. Many methodologies or findings from this field mainly generated into journal for others to take advantages and improve as upcoming research on projects. The methodology refers to the overall approach that our project requires. We need to explain our project briefly, demonstrating that we comprehend the meaning of our approaches. The methods are the tools of data collection, the procedure of our project. The procedures or strategies used to find, select, process, and analyse information about a topic are referred to as methodology. [14]

Shrewd Car Parking System Using Arduino UNO Designing, creating and delivering a main edge stopping innovation is called as Smart stopping.[12] It is a vehicle stopping framework that helps drivers locate an empty spot. Utilizing the Ultrasonic sensors in each stopping space, it identifies the nearness or nonattendance of a vehicle. Brilliant Parking framework is demonstrated as a correct, strong and cost-effective approach to guarantee that street clients know precisely where empty car parking spots are.

A particular procedure or set of procedures demonstrating the issue is massive revision of teaching methodology. In a report or article, the methodology section allows the reader to critically evaluate a study's overall validity and reliability. So, this methodology chapter explains what we did and how we did it.

3.2 Block Diagram of the proposed project

Block Diagram of our proposed system is given below:

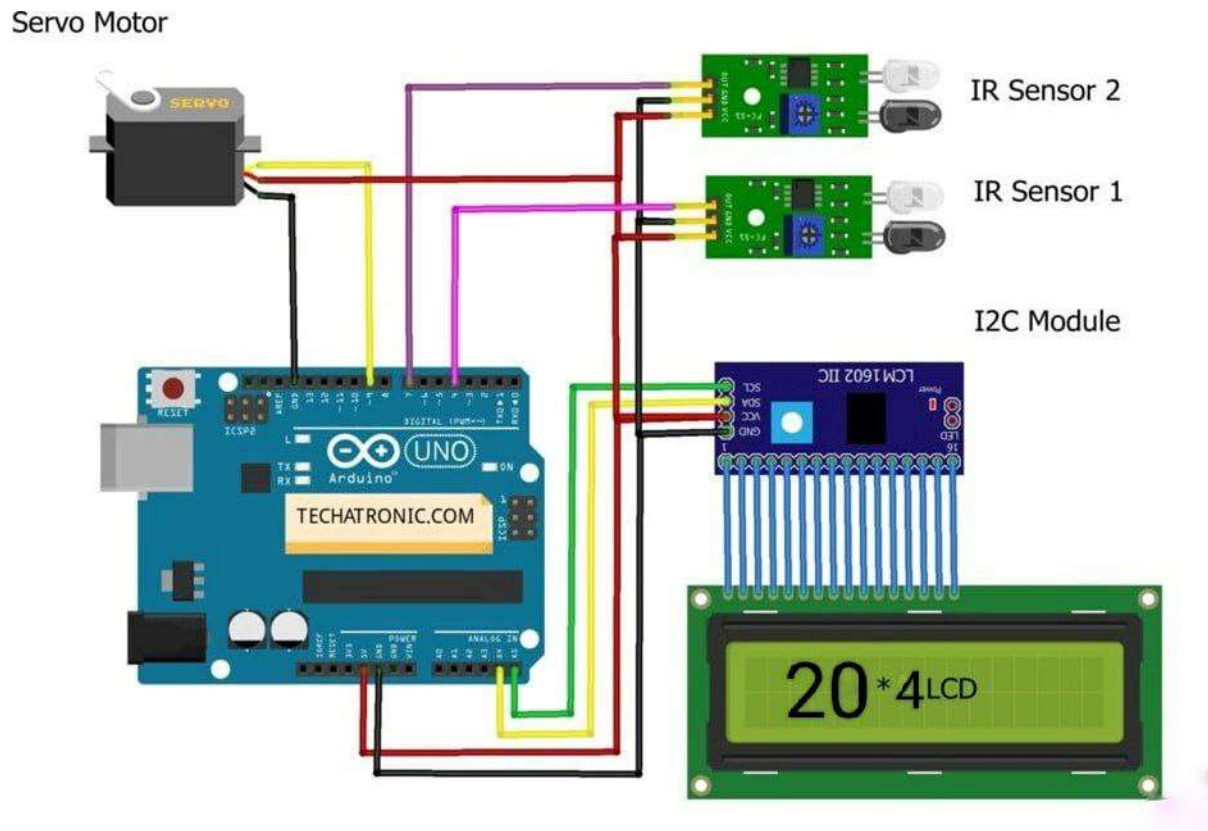


Figure-1: Block diagram of car parking system

The circuit shown in Fig. 1 an Arduino microcontroller (MCU) requires only 5V. As Arduino Uno has an inbuilt 5V voltage regulator, a common 5V supply can be used for the whole system.

The brain of the circuit is Arduino Uno MCU board (BOARD1). It is based on Arduino UNO R3 V1.0 and has 14 digital input/output (I/O) pins, six analogue inputs, servo motor, a USB connection, IR sensor with 12 C module among others. It can be programmed using Arduino IDE software IR sensor are connected to Arduino digital pins 2 and 3 and servo motor are connected to Arduino pins 13, which are used for serial communication.

The LCD display (LCD1) is used to display messages during action. Here, a 20×4 display is used; each character is made of 5×7 dot-matrix. Pins 3, 4, 5 and 6 of the LCD are the control lines connected to preset (PR1) output, pin 12 (Arduino), GND and pin 11 (Arduino). Pins 11,

12, 13 and 14 are data pins of the LCD that are connected to pins 7, 6, 5 and 4 of Arduino, respectively. Preset PR1 is used to adjust the contrast of the LCD display.

3.3 Construction and working principle:

The construction and working of the project are divided into two parts: -

3.3.1) Entry Part:

The project is a sensor based automatic system. The Entry Part of the project consists of Arduino Mega microcontroller to which a servo motor, IR sensor, LCD and ultrasonic sensor are interfaced. The servo motor acts as a gate at the entrance and it opens and closes when the IR sensor detects presence of car. The input IR sensors reads presence of any car and sends to the controller Arduino UNO board. While entry IR sensor detect car, it opens the door in condition of having available slot. After entering the car, the gate gets closed with the help of servo motor. The motor gets instruction from the board. The LCD displays the parking slots which are empty for the car drivers. While A car gets into a slot, the corresponding slot status in the LCD display shows "Have space". At the time of exiting a slot, the status of the slot changed to "Vacant". When all the slots are booked, the system doesn't allow the fourth car to enter, and shows "Sorry parking full". The ultrasonic sensors detect the presence and absence of car in each parking slot.

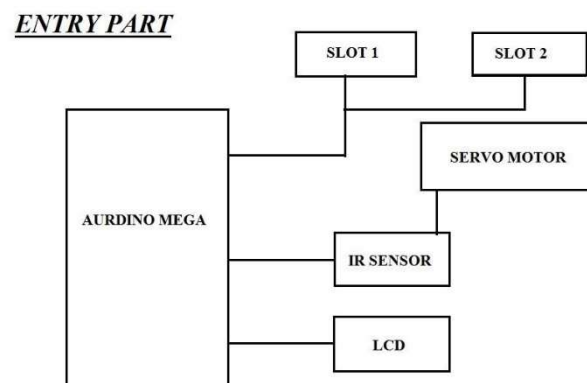


Figure-2: Entry part of car parking system

3.3.2) Exit Part

The Exit Part of the project consists of Arduino Uno to which a servo motor, IR sensor and the object counter are interfaced. The servo motor acts as a gate at the entrance and it opens and closes when the IR sensor detects presence of car. When a car exits from a slot, the exit IR sensor detects the car and opens the gate with the servo motor. Then total available free slot count increase by 1 in the screen.

EXIT PART

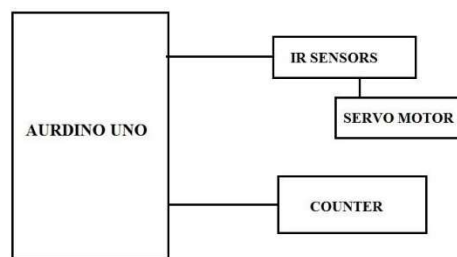


Figure-3: Exit part of car parking system

The system will be very reliable and understandable to the users.

3.3.1 Process of Work:

The project is a difficult technological problem that requires a certain method or working procedure to obtain the required outcomes. Users' cultural, religious, language, morality, and socioeconomic characteristics, such as education level, money, and gender, have no influence on how the process is developed, as explained in this research.

As a result, there is no such conflict in our system. Our technology is cost-effective and simple to deploy, and it will not cause any linguistic difficulties. The religious beliefs of people will be unaffected by this method. In addition, our method will not obstruct gender equity.

Our method will not affect other department without parking. It hasn't any issue with public security and people will be benefitted. Traffic Jam, a major problem of the country will be solved. This approach will have a favourable influence on smart parking and may also be useful to all drivers. As a result, our system will not have any religious, economic, or moral disagreements because it is clarified that the device will save time and labour cost.

We prepared a poll based on our working method to learn about people's perceptions of whether our project would truly cause conflict due to language, religion, or culture. The following are excerpts from the survey:

Does the system have any impact on Religion?

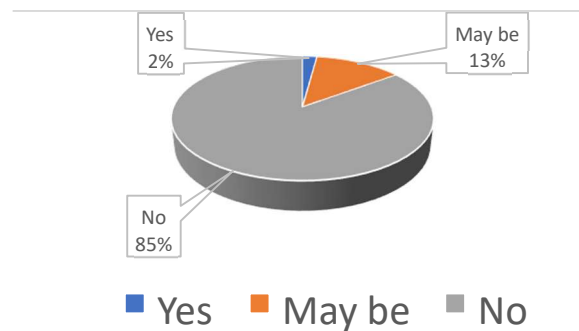


Figure-4:(a)

Can the project create any cultural conflict?

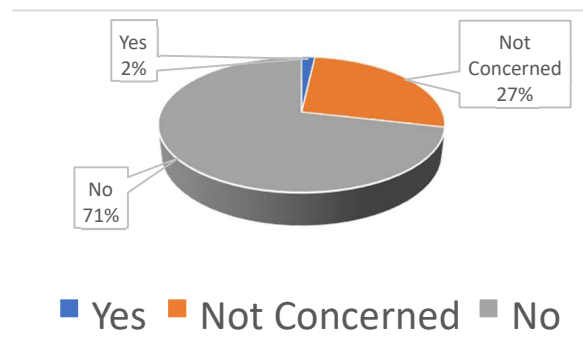


Figure-5: Graphical Representation of car parking system

3.4 Hardware Design

Hardware equipment that we need in order to build the project are given below:

1. Arduino UNO
2. IR Sensor
3. LCD Screen
4. Bread Board
5. Power Supply
6. Connecting Wires
7. 12 C module

3.5 Description of the important component:

First of all, we use Proteus software to complete the simulation. in this simulation, we've used some important components to build the whole parking system, which are Arduino UNO R3 V1.0, LEDGREEN, LM044L 20x4 Alphanumeric Display, MOTOR-PWMSERVO, POT variable resistor, IR sensor with module.

3.5.1 Arduino Board:

Arduino is a project made by the largest technical community of engineers, developers and hobbyists whose goal is to develop ideas and interactive control projects around the world, based on different types of electronic panels but programmed in a language Single programming and free.

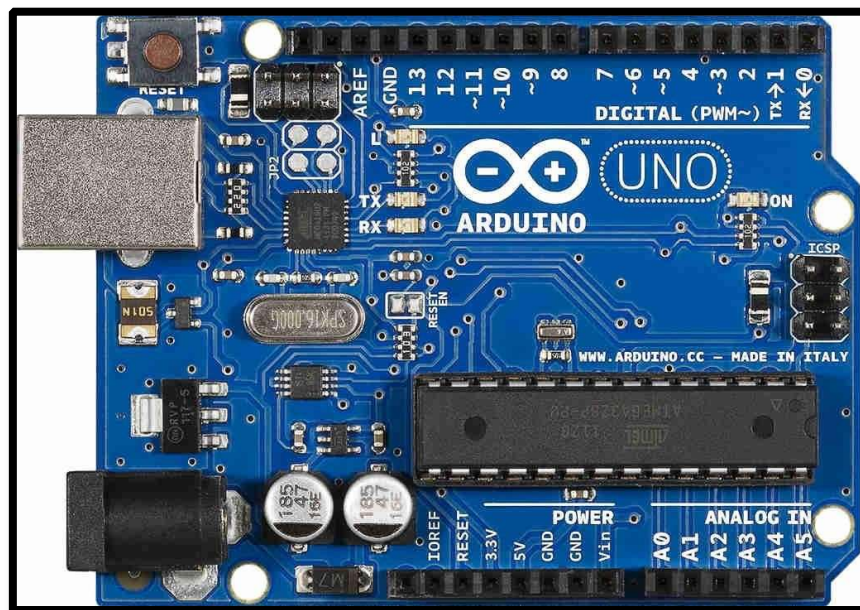


Figure 6: Arduino UNO Board.

Sure, Arduino is not the only electronic controller in the market, but there are many micro-controllers available in the market such as Parallax, Basic Stamp and the most powerful competitor, Raspberry Pi, all with powerful capabilities and the ability to work. Full projects, of course depending on the needs of your project, but what distinguishes the following Arduino Open-source platform Open-Source Hardware and Software.[23]

Arduino is made primarily of ATMEGA8 and ATMEGA168 controllers, and all its designs are licensed under the Creative Commons license. This is the most important feature for electronic circuit designers because it makes it easier for them to design anything they want. The software is written in C ++, and is available to everyone to download and programmers can modify it according to their needs.

It is the most widely used and widely used one of the many Arduinos. It is the first choice for beginners. It is easy to learn. It operates with an ATmega328 controller. This type has 14 digital ports (I / O), 6 of which can be used as ports to control the " PWM Outputs ", the most important feature of this type is that the control chip " ATmega328 "is not fixed in the board, but installed on the holder of the integrated circuit" IC ". This feature makes it the best option for beginners so that if burned the slide while working on your project by mistake, you can Restore your work on the board as soon as you change the slide The ATmega328 controller is similar to the same model.

The first-ever type of Arduino's motherboard is the ATmega32u4 controller, which has a unique feature that contains a built-in USB connection, eliminating the need to use a secondary processor. The feature allows the panel as soon as it is connected to your device to appear as a keyboard and mouse, which makes it ideally suited to build various applications that enable you to control your PC. [23]

Advantages of Arduino:

Simplicity

Arduino's paintings are designed to suit the needs of all engineers, designers, professors, students, and interactive electronics enthusiasts around the world.

The price

The Arduino Plate is less expensive than any competitor of the same type. The most expensive painting is not more than 600.

Self-Assembly

Easy to deal with and easy to connect circuits, as we mentioned in our first article that it was an easy solution to the problem of microcontrollers and complex connections.

Multi-platform

The Arduino program has the ability to work with all the different operating systems of Windows, Mac and Linux, while most other boards running on Windows only

Easy and simple software environment

The "Environment" programming environment is designed to be easy for beginners and powerful professionals and its programming language "Arduino C" is easy to learn.[23]

3.5.2 20x4 Alphanumeric Display

20x4 Alphanumeric Display means it can display 20 characters per line and there are 4 such lines. In this LCD each character is displayed in 5x7 pixel matrix. This LCD has two registers, namely, Command and Data.

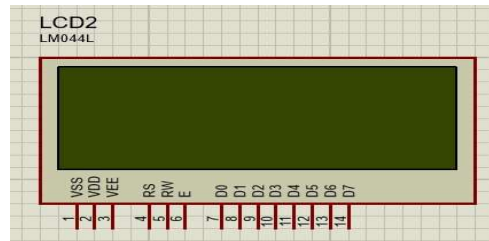


Figure 8: 16x2 Alphanumeric Display

The term LCD stands for liquid crystal display. It is one kind of electronic display module used in an extensive range of applications like various circuits & devices like mobile phones, calculators, computers, TV sets, etc. These displays are mainly preferred for multi-segment light-emitting diodes and seven segments. The main benefits of using this module are inexpensive; simply programmable, animations, and there are no limitations for displaying custom characters, special and even animations etc.[24]

LCD 20X 4 PIN DIGRAM

The 20 x 4 pin LCD pinout is shown below

- VSS (Pin 1): Ground pin of the LCD module.
- VDD (Pin 2): Power supply pin of the LCD module (typically +5V).
- V0 (Pin 3): Contrast adjustment pin. It usually connects to a variable resistor (potentiometer) to adjust the contrast of the display.
- RS (Pin 4): Register Select pin. Selects command register when low, and data register when high.
- RW (Pin 5): Read/Write pin. Selects read mode when high, and write mode when low.
- E (Pin 6): Enable pin. Initiates data read/write when high to low transition occurs.
- D0 (Pin 7): Data pin 0 (used in 8-bit mode only).

- D1 (Pin 8): Data pin 1 (used in 8-bit mode only).
- D2 (Pin 9): Data pin 2 (used in 8-bit mode only).
- D3 (Pin 10): Data pin 3 (used in 8-bit mode only).
- D4 (Pin 11): Data pin 4. (In 4-bit mode, this is the first data pin).
- D5 (Pin 12): Data pin 5.
- D6 (Pin 13): Data pin 6.
- D7 (Pin 14): Data pin 7.
- A (Pin 15): Anode of the LED backlight.
- K (Pin 16): Cathode of the LED backlight.

Common Pin Connections:

- VSS to Ground.
- VDD to +5V.
- V0 to the middle pin of a 10k potentiometer, with the other two pins of the potentiometer connected to VDD and Ground to adjust the contrast.
- RS to a digital pin on the microcontroller.
- RW typically connected to Ground (for write mode).
- E to a digital pin on the microcontroller.
- D4-D7 connected to digital pins on the microcontroller (if using 4-bit mode).
- A through a current-limiting resistor (typically 220Ω) to +5V.
- K to Ground.

Features of LCD 20x4

- 20 Characters per Line, 4 Lines: Allows for a total of 80 characters display.
- HD44780 Controller: Standardized interface compatible with many microcontrollers.
- 4-bit or 8-bit Data Interface: Flexible data bus options.
- LED Backlight: Provides visibility in low-light conditions with adjustable brightness.
- Adjustable Contrast: Fine-tune contrast using a potentiometer.
- +5V Operating Voltage: Typical power requirement with low power consumption.
- Built-in and Custom Character Set: Supports alphanumeric characters and up to 8 custom characters.
- Wide Software Support and Documentation: Easy integration with extensive libraries and tutorials.

3.5.3 SERVO MOTOR:

- A Servo Motor is a small device that has an output shaft. This shaft can be positioned to specific angular positions by sending the servo a coded signal. As long as the coded signal exists on the input line, the servo will maintain the angular position of the shaft.



Figure 9: SERVO MOTOR

- The servo engine is a motor that comes with a Gear gearbox and a Shaft transmission that gives motion greater torque and greater precision. This engine can rotate 180 degrees and in some types 360 degrees. The servomotor is internally made up of a "mostly microcontroller"

control circuit. When the engine gives pulses at a certain time constant, the engine rotates to the angle according to that time constant. Edited with the trial version of To remove this notice, visit www.pdfpro.co 14 In each type, the time constant varies from one engine to another according to the manufacturer and the technical bullet in that comes with the servo engine. In the Arduino, programming environment there is a library called Servo Library installed in the program. This library gives us the ability to control most of the 180-degree Cervo drives. At the end of this post, you will have the ability to use the library's commands through practical examples.[25]

3.5.3.1 Servo Motor Working Mechanism

It consists of three parts:

1. Controlled device
2. Output sensor
3. Feedback system

It is a closed-loop system where it uses a positive feedback system to control motion and the final position of the shaft. Here the device is controlled by a feedback signal generated by comparing output signal and reference input signal.

Here reference input signal is compared to the reference output signal and the third signal is produced by the feedback system. And this third signal acts as an input signal to the control the device. This signal is present as long as the feedback signal is generated or there is a difference between the reference input signal and reference output signal. So, the main task of servomechanism is to maintain the output of a system at the desired value at presence of noises.[25]

3.5.4 POT variable resistor:

The potentiometer (POT) is the most common variable resistor. It functions as a resistive divider and is typically used to generate a voltage signal depending on the position of the potentiometer.

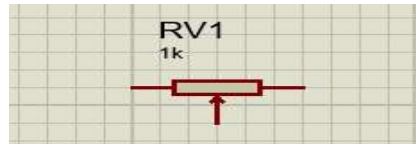


Figure 10: POT variable resistor

A potentiometer is a manually adjustable variable resistor with 3 terminals. Two of the terminals are connected to the opposite ends of a resistive element, and the third terminal connects to a sliding contact, called a wiper, moving over the resistive element. The potentiometer essentially functions as a variable resistance divider. The resistive element can be seen as two resistors in series (the total potentiometer resistance), where the wiper position determines the resistance ratio of the first resistor to the second resistor. If a reference voltage is applied across the end terminals, the position of the wiper determines the output voltage of the potentiometer.[26]

3.5.5 IR Sensor:

An infrared (IR) sensor is a type of electrical gadget that detects and measures infrared radiation in its surroundings. In the year 1800, an astronomer named William Herchel made an unintentional discovery of infrared radiation. He found that the temperature just beyond the red light was the highest when measuring the temperature of each hue of light (separated by a prism). Because the wavelength of IR is longer than that of visible light, it is invisible to the human eye (though it is still on the same electromagnetic spectrum). Infrared radiation is emitted by everything that emits heat (anything with a temperature over roughly five degrees Kelvin).[27]



Figure 11: Obstacle Infrared Sensor module MOD56

The IR sensor module includes five essential parts like IR Tx, Rx, Operational amplifier, trimmer pot (variable resistor) & output LED. The pin configuration of the IR sensor module is discussed below.

- VCC Pin is power supply input
- GND Pin is power supply ground
- OUT is an active-high o/p

The main specifications and features of the IR sensor module include the following.

- The operating voltage is 5VDC
- I/O pins – 3.3V & 5V
- Mounting hole
- The range is up to 20 centimeters
- The supply current is 20mA
- The range of sensing is adjustable
- Fixed ambient light sensor

Types of IR Sensor

The classification of IR sensors can be done based on the application which includes the following.

- Active Infrared Sensors
- Passive Infrared Sensors

Active IR Sensor

This type of sensor includes both the emitter & the receiver which are also known as transmitter & receiver. In most situations, a laser diode or LED is used as a source. For non-imaging infrared sensors, LED is used whereas laser diode is used for imaging infrared sensors.

The working of an infrared sensor can be done through radiating energy, detected and received through the detector. Further, it is processed through a signal processor to fetch the required data. The best examples of active infrared sensors are reflectance & break beam sensors.[27]

Passive Infrared Sensor

Passive Infrared Sensor (PIR) includes detectors only and this kind of sensor uses targets like infrared transmitters or sources. Here, the object will radiate the energy & detects it through infrared receivers. After that, a signal processor is used to understand the signal to obtain the required data. The best examples of PIR sensors are bolometer, Pyro-Electric Detector, Thermocouple-Thermopile, etc. PIR sensors are available in two types like thermal IR sensor and quantum IR sensor.[27]

3.5.6 I2C MODULE

I2C (Inter-Integrated Circuit) is a serial communication protocol that allows multiple devices to communicate with each other using only two wires: a clock line (SCL) and a data line (SDA). An I2C module typically consists of an IC that acts as an interface between the microcontroller and other devices such as sensors, displays, EEPROMs, etc., using the I2C protocol. It simplifies the connection of multiple devices to a microcontroller, as it requires fewer pins compared to other communication protocols

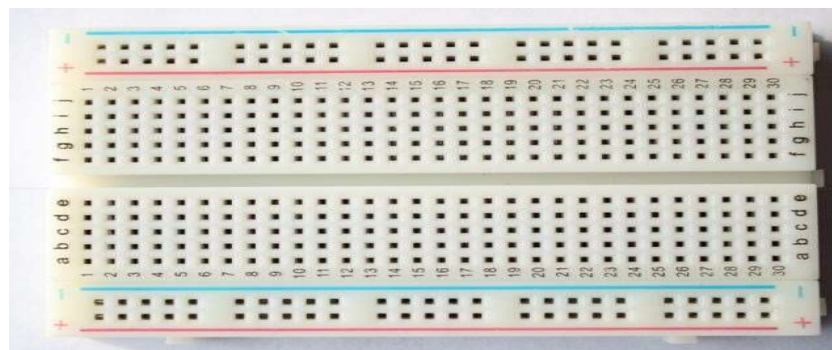


FIG 12 : i2C MODULE

3.5.7 .Other Components:

The project uses some other electronic components like:

- Board:



- Connecting wires:



(b)

Figure -13: Other Components.

3.6 Flow Chart

The flow chart includes how the system works. The program flow chart is given below:

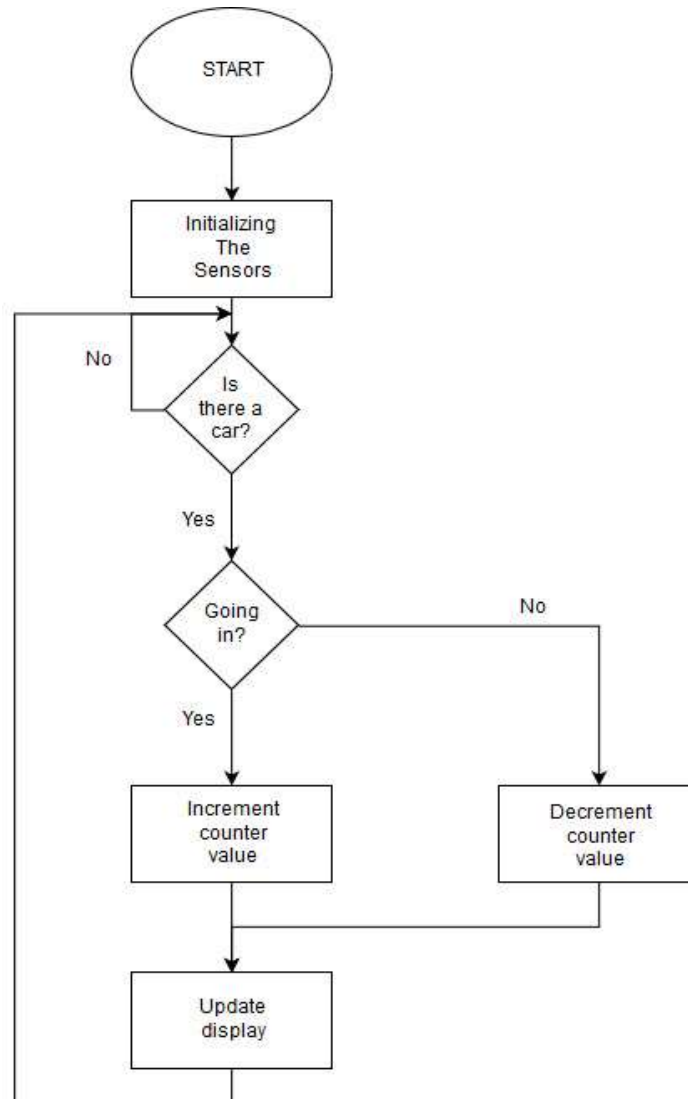


Figure-14: Flow chart of Smart Car Parking System

3.7 Implementation:

To implement the Smart Parking System, we have used Proteus professional 8 simulating software and Arduino IDE. Figure 9 shows the simulation implementation of the project.

Firstly, A new project created without any firmware in proteus. After creating the blank new project, there will be a schematic diagram window. In this schematic window, the simulation circuit is constructed. All of the devices used here, were found under pick device option. Arduino UNO R3 V1.0 was chosen as microcontroller board for Smart Parking System. This system works on the basis of some major components such as Arduino UNO R3, POT variable resistor, LED-GREEN, IR-SENSOR, MOTOR-PWMSERVO, LM044L 16x2 Alphanumeric Display

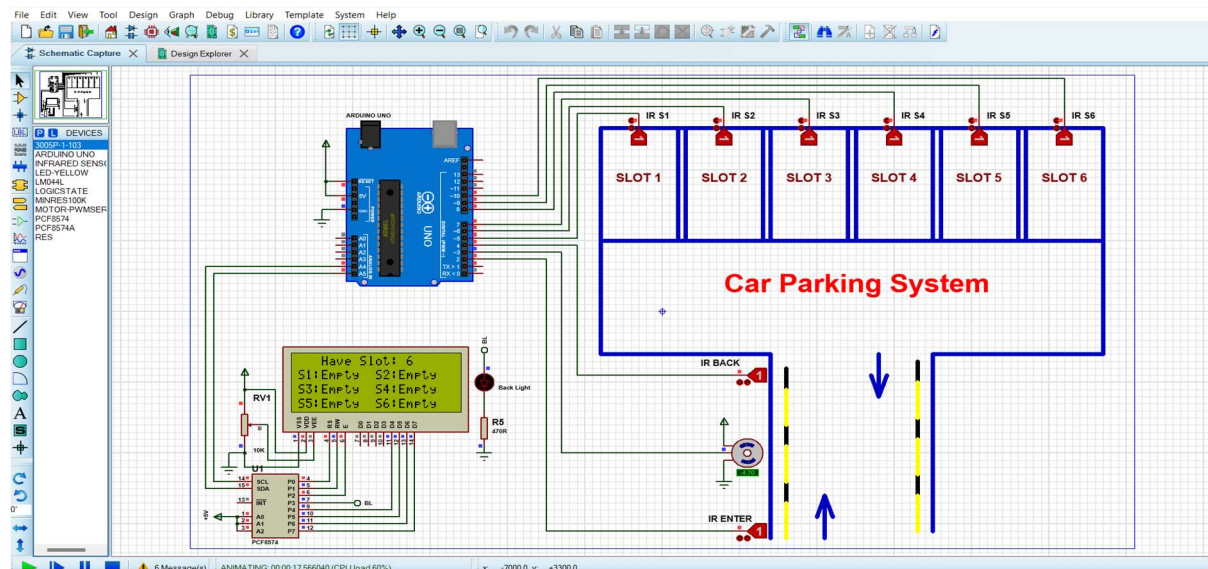
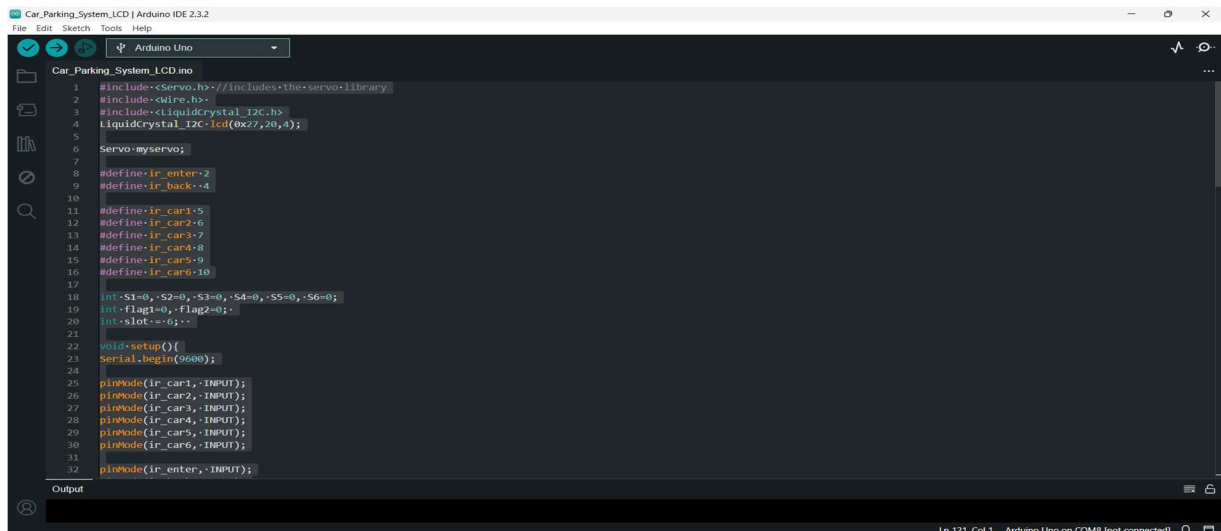


Figure 15: The design of Smart Parking System using Arduino UNO

had used in this project. The display was connected to the board through the expander. So basically, we have used the LM044L 16x2 Alphanumeric Display to print the slot list, each slot payment status of the parking lot. For the connection of the Display, we used Remote 8-bit I/O expander to reduce the connection with the Arduino pin. As it is a simulation-based project, here IR sensor use was not too easy. The logic states used to show the functionalities of the IR sensors. Every logic states connected to individual digital I/O pin of the Arduino UNO. The servo motor was



```
1 #include<Servo.h> //includes the servo library
2 #include<Wire.h>
3 #include<LiquidCrystal_I2C.h>
4 LiquidCrystal_I2C lcd(0x27,20,4);
5
6 Servo myservo;
7
8 #define ir_enter 2
9 #define ir_back 4
10
11 #define ir_car1 5
12 #define ir_car2 6
13 #define ir_car3 7
14 #define ir_car4 8
15 #define ir_car5 9
16 #define ir_car6 10
17
18 int s1=0, s2=0, s3=0, s4=0, s5=0, s6=0;
19 int flag1=0, flag2=0;
20 int slot=6;
21
22 void setup() {
23   Serial.begin(9600);
24
25   pinMode(ir_car1, INPUT);
26   pinMode(ir_car2, INPUT);
27   pinMode(ir_car3, INPUT);
28   pinMode(ir_car4, INPUT);
29   pinMode(ir_car5, INPUT);
30   pinMode(ir_car6, INPUT);
31
32   pinMode(ir_enter, INPUT);
```

Figure16: The code window on Arduino IDE

also connected with a digital pin for get instruction and it get external power to run. Two IR sensor set at the gate of the parking spot for detect entry and exit of car. Each slot also has IR sensor to identify slot is empty or not. The servo motor rotates 180 degrees to open the gate and 90 degrees to close the gate. The green LED lights while the whole system is turned on. Pot variable resistor ensures different power supply to the LCD display. After setting up the components as figure 9, we went to Arduino IDE and installed all the necessary libraries to run the project successfully. Then the codes/commands were written and uploaded to the Arduino Board. After upload finely, the code worked without any errors.

3.8 Test/Experimental setup:

While doing the setup, the project was evaluated that every functionality is accomplished or not. The code was edited as proposed requirements.

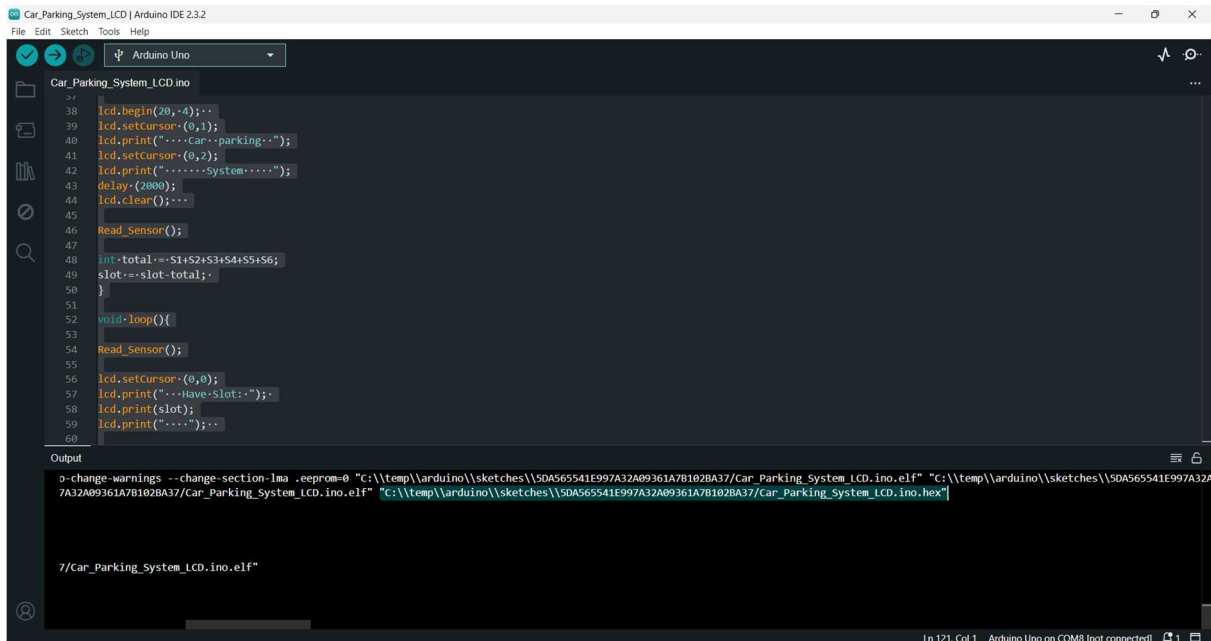


Figure-17: Hex File in Arduino IDE

After running the instruction code finely, there will be a .HEX file found under IDE's compiler console and this file location should be copied. In proteus, double click in the Arduino board and in the program file section, paste the link without any change. Click ok and run the simulation. The system will run expectedly. All the output was observed while running the simulation as like following.

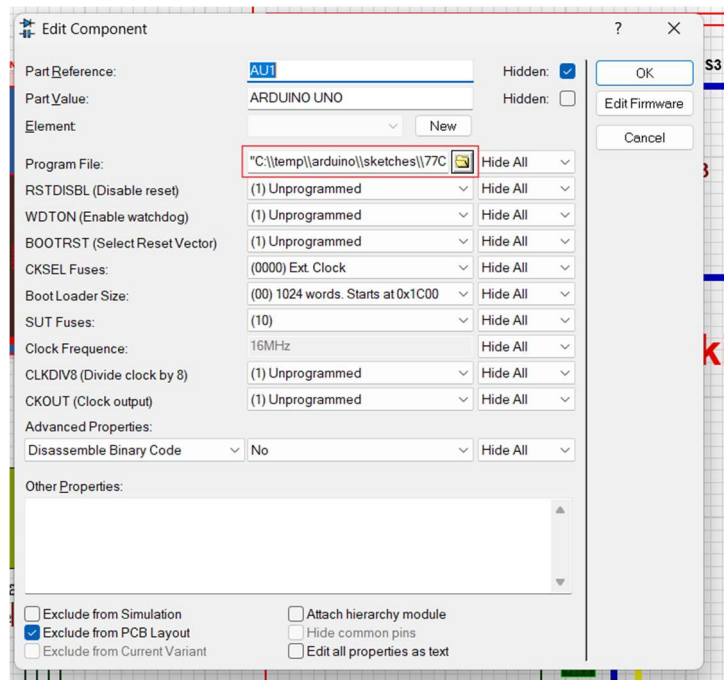


Figure-17: The program file location for Arduino

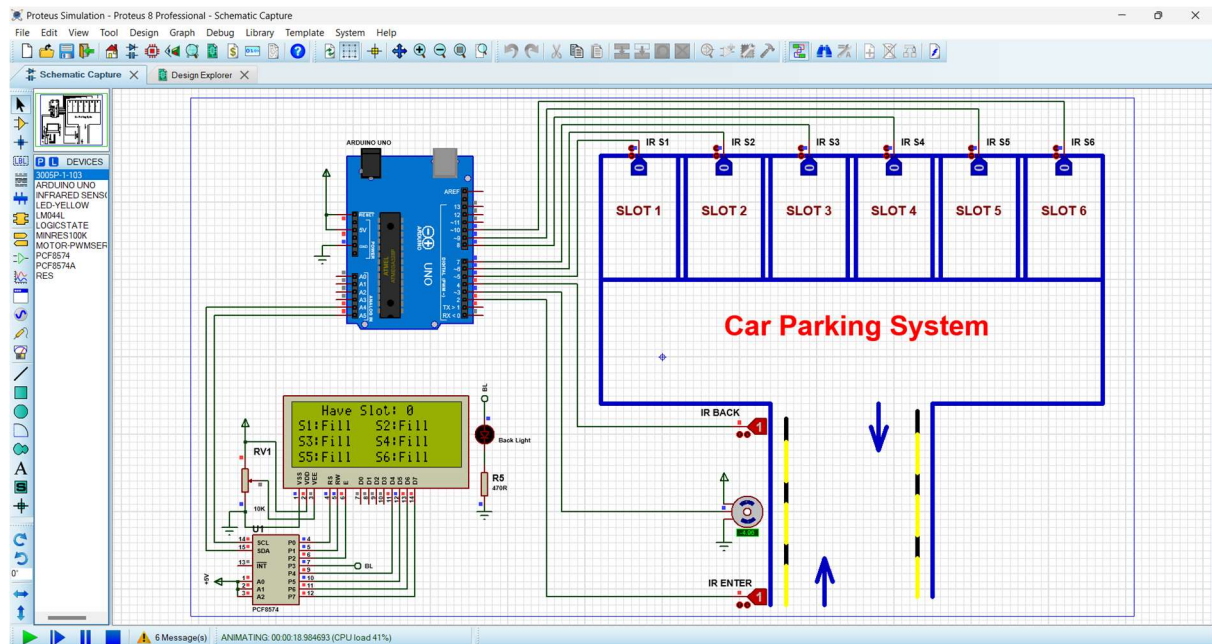


Figure-19: After Simulation Run Successfully

4. Hardware Implementation

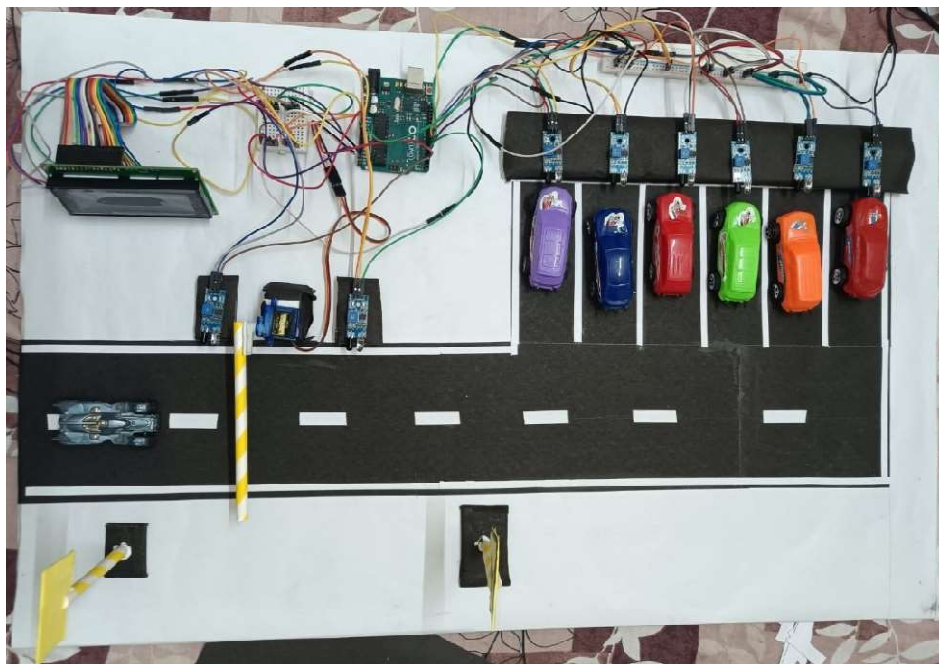
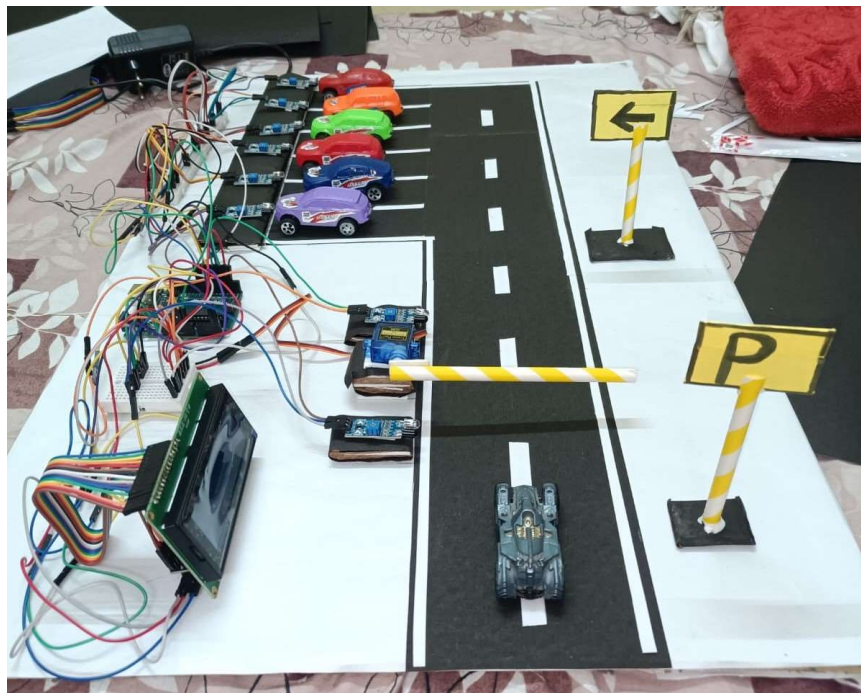


Fig 19 : Hardware implementation

5. RESULTS AND DISCUSSION

1. It guarantees snappy and computerized parking and simple recovery of vehicles.
2. Up to 6 cars can be effectively and securely parked in the outlined model.
3. The surface space required is identical to the parking spot of two cars as it were.
4. Most reasonable for parking in workplaces, shopping centres and comparable spots.
5. Low support levels are required by the framework.
6. Sensors utilized have high affectability and are anything but difficult to deal with.
7. Minimal effort framework, giving most extreme computerization.
8. It doesn't require observable pathway operation.
9. Cordial reorientation of cars for driving in and out.
10. Security of vehicle.

6. REFERENCES

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7.CONCLUSION

Our project detects the empty slots and helps the drivers to find parking space in unfamiliar city. The average waiting time of users for parking their vehicles is effectively reduced in this system. The optimal solution is provided by the proposed system, where most of the vehicles find a free parking space successfully. Our preliminary test results show that the performance of the Arduino UNO based system can effectively satisfy the needs and requirements of existing car parking hassles thereby minimizing the time consumed to find vacant parking lot and real time information rendering. This smart parking system provides better performance, low cost and efficient large scale parking system. When car enters the parking area, the driver will park the car in the nearest empty slot when slot is occupied the LED light glows and when slot is empty LED lights are turned off chromatically indicating that the parking slot is empty to be occupied. It also eliminates unnecessary travelling of vehicles across the filled parking slots in a city. Smart Parking solutions are designed to provide drivers an ultimate solution on their journey from the beginning to end without searching for parking, cost, travel time etc. This advantage comes by paying marginal fees to the smart parking service providers. To change a culture which has been existing for several centuries is a humongous task. Parking has always been an at the moment affair with direct cash exchange. The inclusion of technology in this method is a change in culture which will take the time to establish. Smart Parking is one of the most adopted and fastest growing smart city solutions across the world. Airports, universities, shopping centres and city garages are just a few entities that have begun to realize the significant benefits of automated parking technology.